In [ ]:

In [2]:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

%matplotlib inline

sns.set_theme(style="whitegrid")
```

In [3]:

```
df=pd.read_csv("train.csv")
df.head()
```

Out[3]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	I
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0

In [4]:

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype  
---  ------       --------------  -----  
 0   PassengerId  891 non-null    int64  
 1   Survived     891 non-null    int64  
 2   Pclass       891 non-null    int64  
 3   Name         891 non-null    object 
 4   Sex          891 non-null    object 
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64  
 7   Parch        891 non-null    int64  
 8   Ticket       891 non-null    object 
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object 
 11  Embarked     889 non-null    object 
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

In [5]:
```
df.describe()
```

Out[5]:

	PassengerId	Survived	Pclass	Age	SibSp	Parch	
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.00(
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.00(
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.91(
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.00(
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329

In [6]:
```
df["Sex"].value_counts()
```

Out[6]:
```
Sex
male      577
female    314
Name: count, dtype: int64
```

In [7]:
```
df["Pclass"].value_counts()
```

Out[7]:
```
Pclass
3    491
1    216
2    184
Name: count, dtype: int64
```

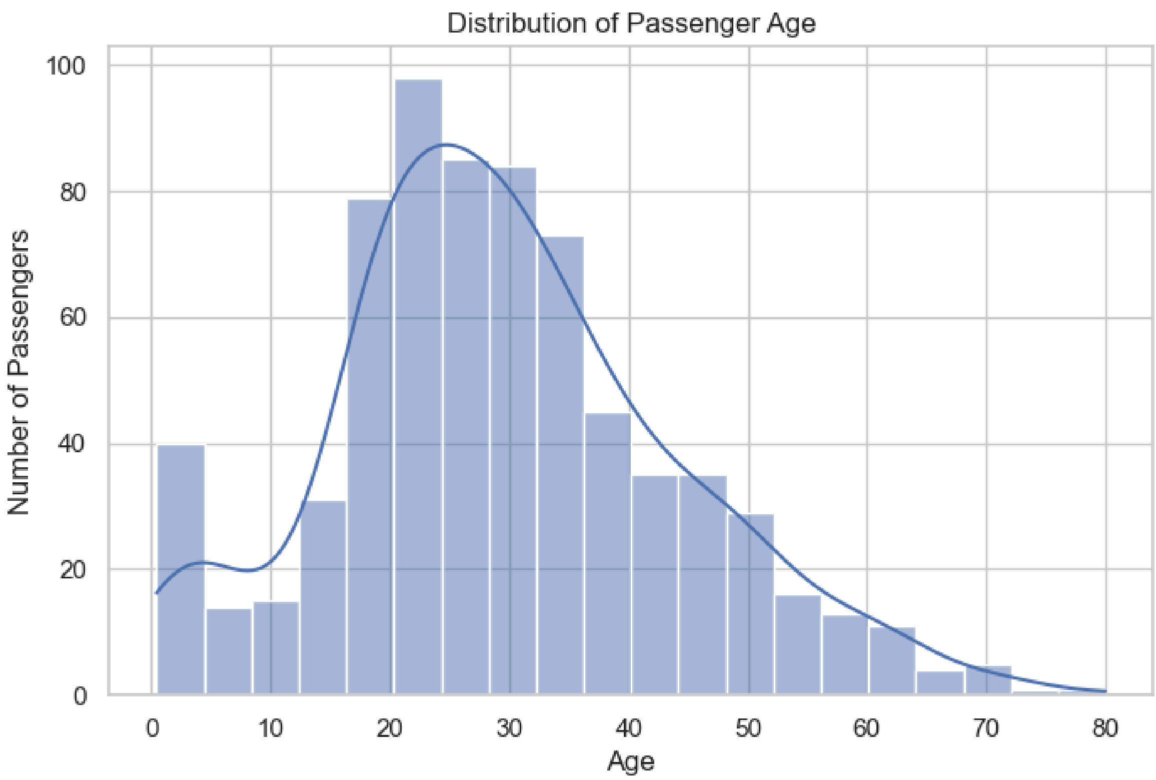
In [8]:
```
df["Embarked"].value_counts()
```

Out[8]:
```
Embarked
S    644
C    168
Q     77
Name: count, dtype: int64
```

In [9]:
```
df.isnull().sum()
```

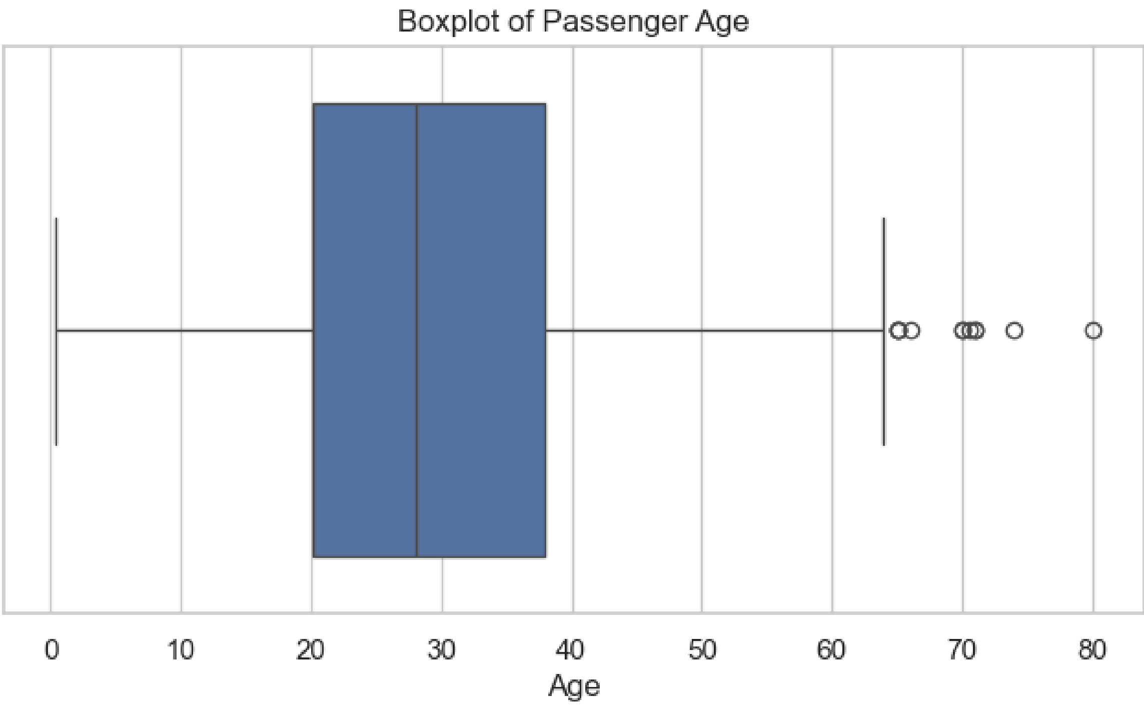
Out[9]:
```
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
```

In [10]:
```
plt.figure(figsize=(8, 5))

sns.histplot(data=df, x="Age", bins=20, kde=True)

plt.title("Distribution of Passenger Age")
plt.xlabel("Age")
plt.ylabel("Number of Passengers")

plt.show()
```

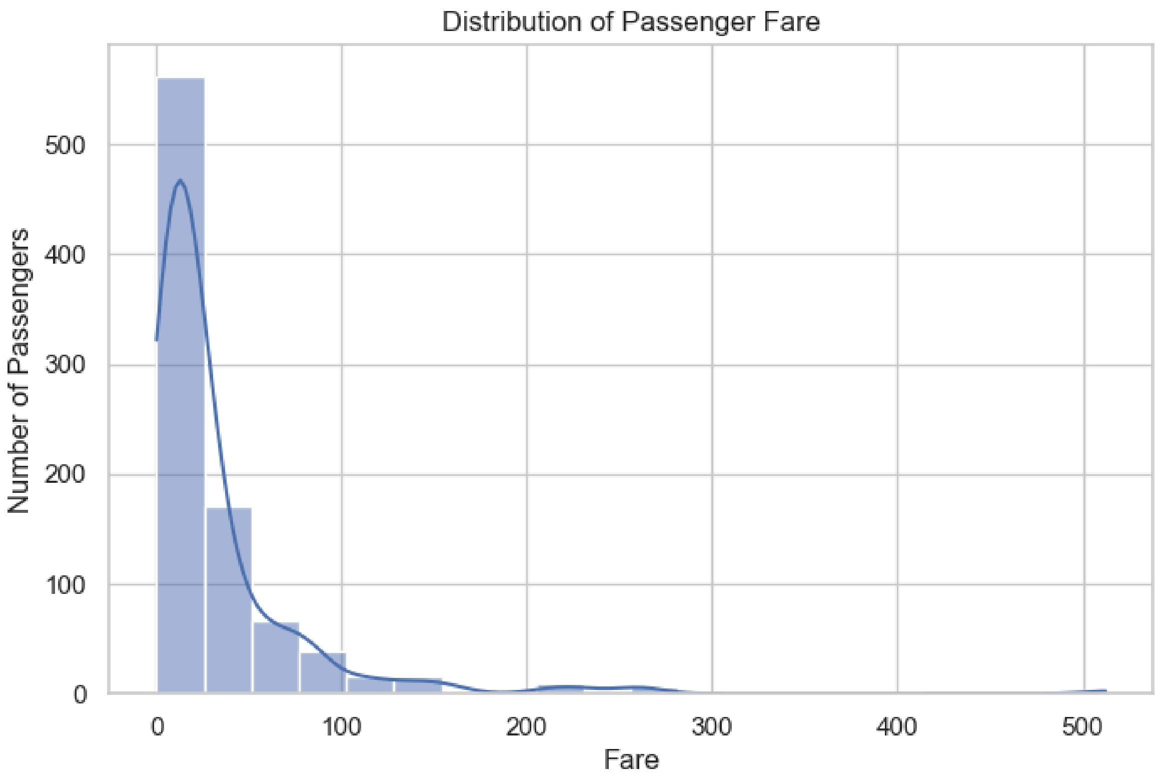


In [11]:
```
plt.figure(figsize=(8, 4))

sns.boxplot(x=df["Age"])
```

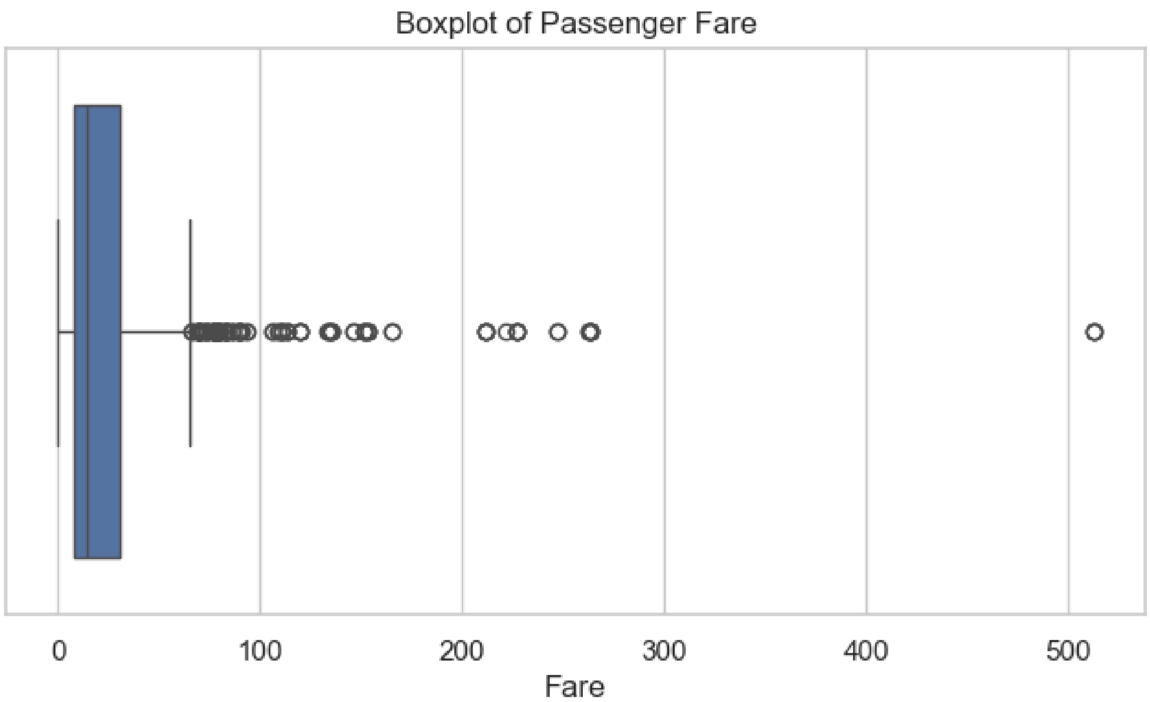
```python
plt.title("Boxplot of Passenger Age")
plt.xlabel("Age")

plt.show()
```



In [12]:
```python
plt.figure(figsize=(8, 5))

sns.histplot(data=df, x="Fare", bins=20, kde=True)

plt.title("Distribution of Passenger Fare")
plt.xlabel("Fare")
plt.ylabel("Number of Passengers")

plt.show()
```

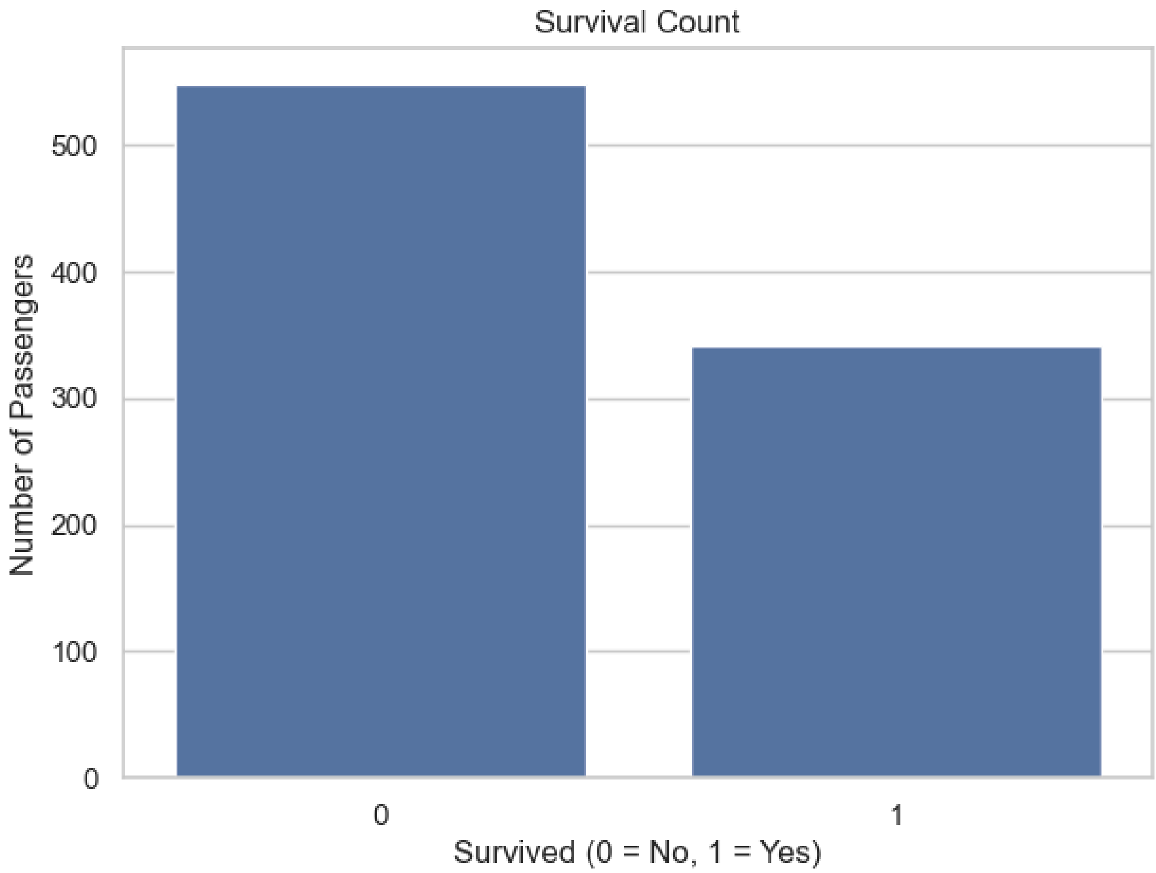


In [13]:
```
plt.figure(figsize=(8, 4))

sns.boxplot(x=df["Fare"])

plt.title("Boxplot of Passenger Fare")
plt.xlabel("Fare")

plt.show()
```



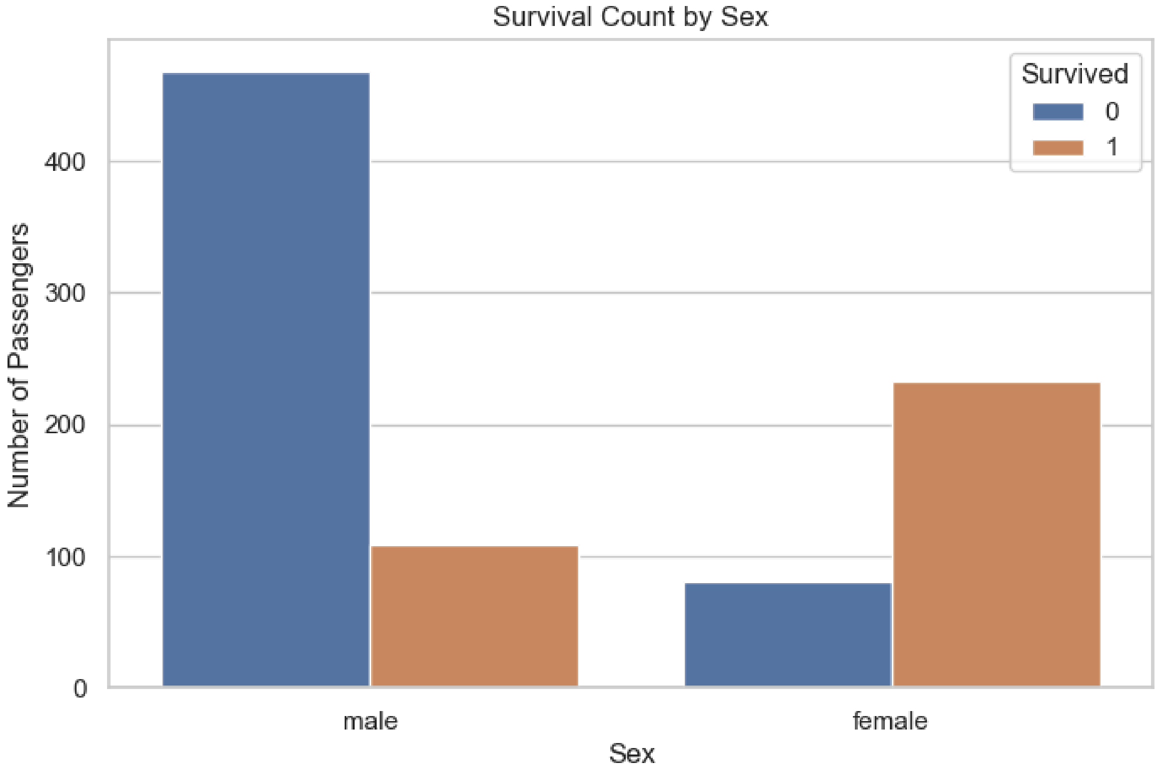
In [14]:
```
plt.figure(figsize=(7, 5))

sns.countplot(data=df, x="Survived")
```

```
plt.title("Survival Count")
plt.xlabel("Survived (0 = No, 1 = Yes)")
plt.ylabel("Number of Passengers")

plt.show()
```

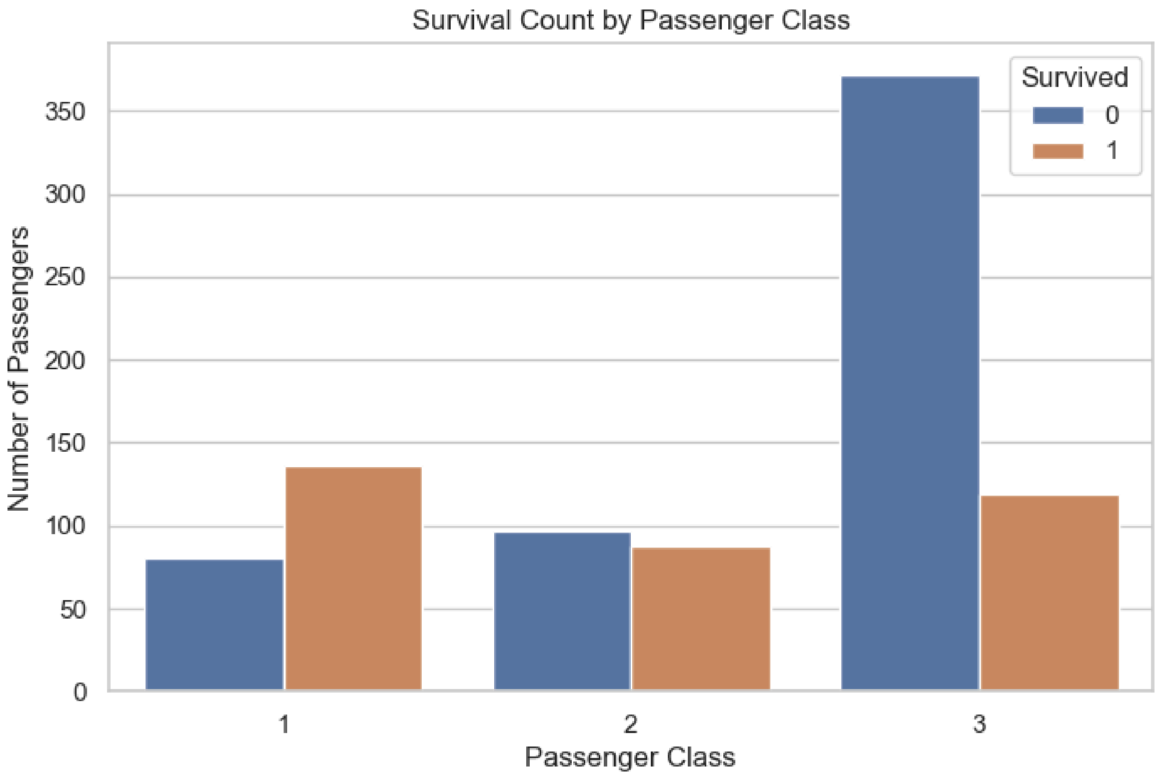


In [15]:

```
plt.figure(figsize=(8, 5))

sns.countplot(data=df, x="Sex",hue="Survived")

plt.title("Survival Count by Sex")
plt.xlabel("Sex")
plt.ylabel("Number of Passengers")

plt.show()
```

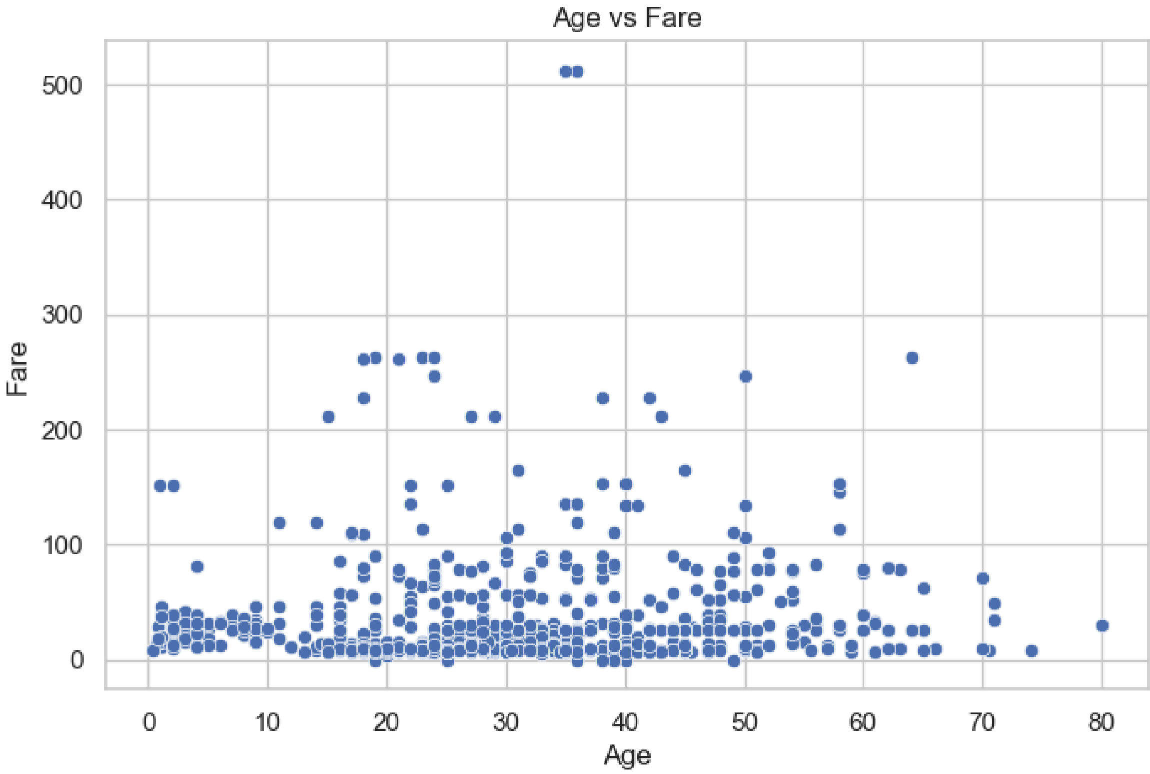


In [16]:
```
plt.figure(figsize=(8, 5))

sns.countplot(data=df, x="Pclass",hue="Survived")

plt.title("Survival Count by Passenger Class")
plt.xlabel("Passenger Class")
plt.ylabel("Number of Passengers")

plt.show()
```



In [17]:

```
plt.figure(figsize=(8, 5))

sns.scatterplot(data=df, x="Age", y="Fare")

plt.title("Age vs Fare")
plt.xlabel("Age")
plt.ylabel("Fare")

plt.show()
```



In [18]:

```
coorelation=df.corr(numeric_only=True)
coorelation
```

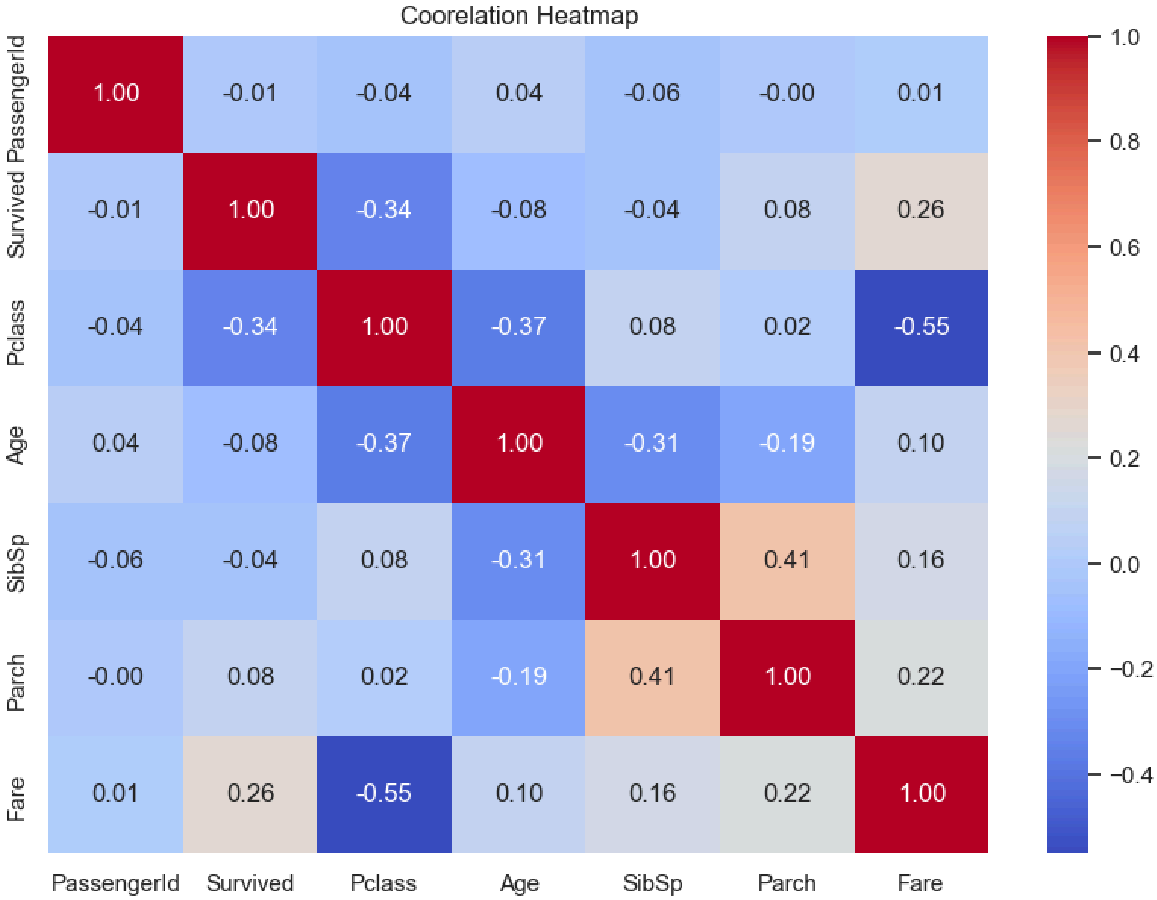
Out[18]:

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fa
PassengerId	1.000000	-0.005007	-0.035144	0.036847	-0.057527	-0.001652	0.01265
Survived	-0.005007	1.000000	-0.338481	-0.077221	-0.035322	0.081629	0.25730
Pclass	-0.035144	-0.338481	1.000000	-0.369226	0.083081	0.018443	-0.54950
Age	0.036847	-0.077221	-0.369226	1.000000	-0.308247	-0.189119	0.09606
SibSp	-0.057527	-0.035322	0.083081	-0.308247	1.000000	0.414838	0.15965
Parch	-0.001652	0.081629	0.018443	-0.189119	0.414838	1.000000	0.21622
Fare	0.012658	0.257307	-0.549500	0.096067	0.159651	0.216225	1.00000

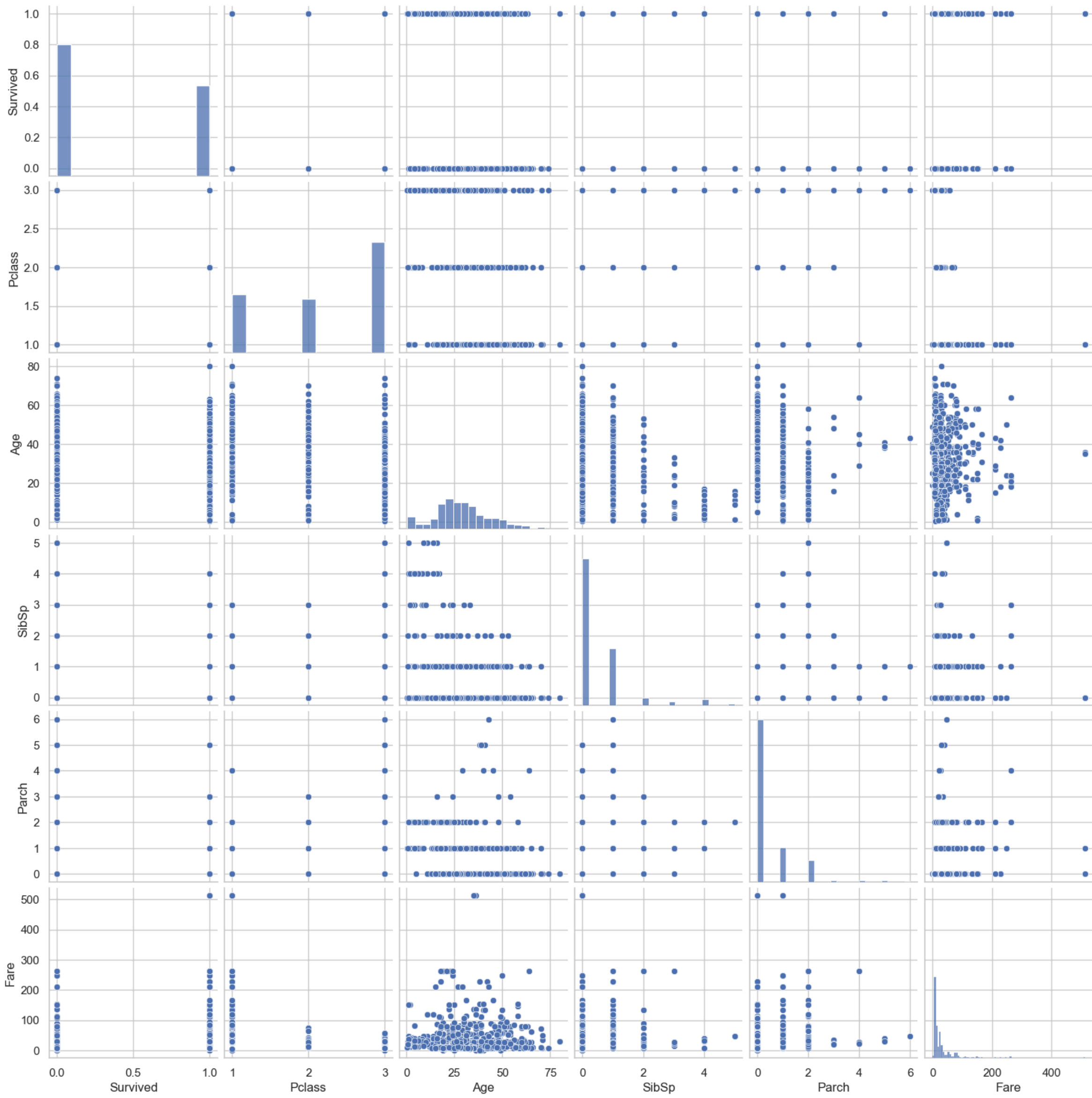
In [19]:

```
plt.figure(figsize=(10, 7))

sns.heatmap(coorelation, annot=True, cmap="coolwarm", fmt=".2f")

plt.title("Coorelation Heatmap")
```

```
plt.show()
```



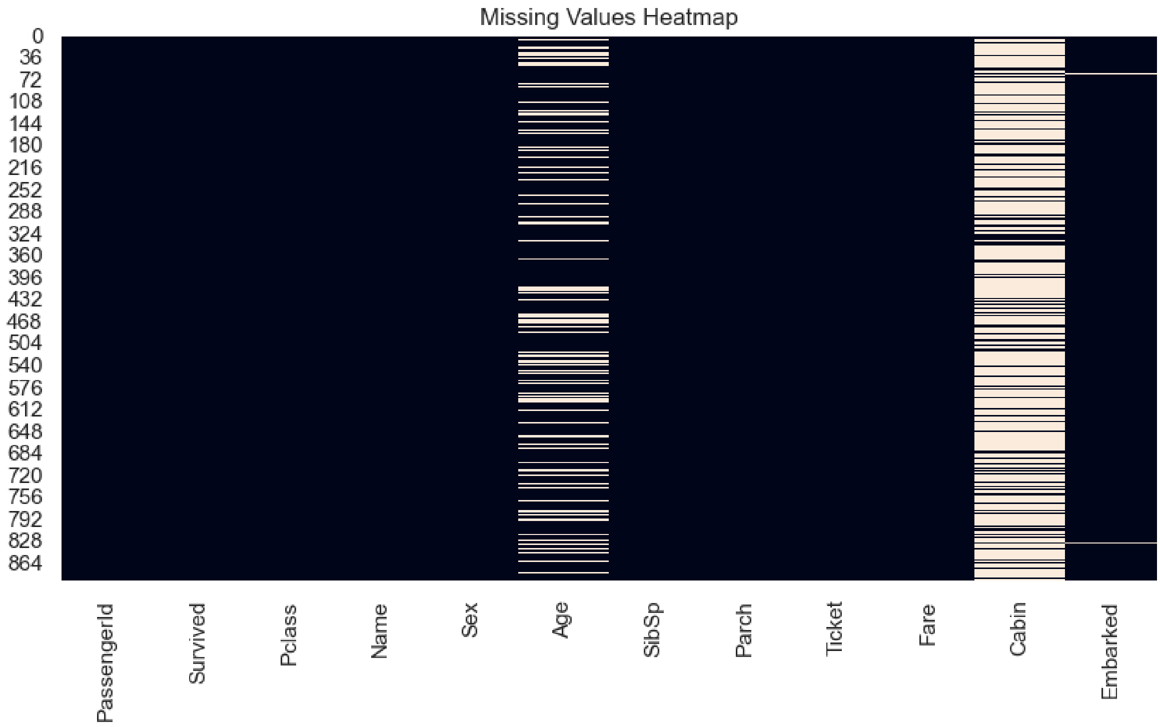
In [20]:
```
sns.pairplot(
    df[["Survived", "Pclass", "Age", "SibSp", "Parch", "Fare"]].dropna()
)

plt.show()
```



In [22]:
```
df.isnull().sum()
```

Out[22]:
```
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
```

In [23]:
```
plt.figure(figsize=(10, 5))

sns.heatmap(df.isnull(), cbar=False)

plt.title("Missing Values Heatmap")

plt.show()
```



In [25]:
```
df["Age"] = df["Age"].fillna(df["Age"].median())
```

In [27]:
```
df["Age"].isnull().sum()
```

Out[27]: 0

In [30]:
```
df["Cabin"] = df["Cabin"].fillna("Unknown")
```

In [31]:
```
df["Cabin"].isnull().sum()
```

Out[31]: 0

In [33]:
```
df["Embarked"] = df["Embarked"].fillna(df["Embarked"].mode()[0])
```

In [34]:
```
df["Embarked"].isnull().sum()
```

Out[34]: 0

In [ ]:

Key Findings

1. The Titanic dataset contains 891 passengers and 12 columns.
2. The overall survival rate was approximately 38%, while around 62% of passengers did not survive.
3. Female passengers had a much higher survival rate than male passengers.
4. Passengers in 1st class generally had better survival outcomes than passengers in 2nd and 3rd class.

5. Most passengers were young or middle-aged, with the median age around 28 years.
6. Fare was strongly right-skewed, with most passengers paying relatively low fares and a small number paying very high fares.
7. Passenger class and fare showed a moderate negative correlation of approximately -0.55.
8. SibSp and Parch showed a moderate positive correlation of approximately 0.41.
9. Age and fare did not show a strong visible relationship in the scatter plot.
10. Missing values in Age, Cabin, and Embarked were handled using Median,"Unknown", and mode imputation respectively.